

longops

Long operations and detailed calculations.
Opérations posées et calculs détaillés.

v0.2.0

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MIT

About

As a mathematics teacher, I sometimes need to show how to carry out arithmetic “by hand”. I am now making available the functions that I originally created for my own use.

I previously used the LaTeX package **xlop**, and since Typst is much more convenient to use, I decided to create my own equivalent. This package is **not** a direct translation of **xlop**: it includes some features that **xlop** does not, while omitting others, according to my own needs.

I occasionally used AI tools (mainly Gemini for the **Order of Operations** module, and also to add features to the other functions that I had initially implemented for just one of them).

Acknowledgments : I wanted to thank [ObaulG](#), the part “durations” is mostly his work.

Functions for handwritten arithmetic layouts ...

- `add-en-ligne()`
- `addition()`
- `base-to-base()`
- `division()`
- `egalite-decimale()`
- `egalite-euclidienne()`
- `from-base()`
- `multiplication()`
- `racine()`
- `sous-en-ligne()`
- `soustraction()`
- `to-base()`

add-en-ligne

Compute the sum of all the supplied numbers inline.

```
$#add-en-ligne(198.7,120,35)$
```

```
198.7 + 120 + 35 = 353.7
```

Parameters

```
add-en-ligne(..ops: arguments) -> content
```

..ops arguments

list of numbers

addition

Parameters

```
addition(  
  base: int,  
  show-base: boolean,  
  show-carry: boolean,  
  carries-color: color,  
  size: length relative length,  
  fr: boolean,  
  hide-result: boolean,  
  border-color: color,  
  solution-color: color,  
  type-mask: str,  
  list: array,  
  solution: boolean,  
  sign: boolean content,  
  space: length relative length fraction,  
  text-size: length relative length,  
  ..args: array arguments  
) -> content
```

base `int`

Base (from 2 to 36)

Default: `10`

show-base `boolean`

to show the base of the numbers and operation

Default: `false`

show-carry `boolean`

show or hide the carries

Default: `true`

carries-color `color`

color of the carries

Default: red

size `length` or `relative length`

Size of the columns

Default: `1em`

fr `boolean`

french decimal separaro “,” or not

Default: `true`

hide-result `boolean`

Hide the carry row and the result

Default: `false`

border-color `color`

Color of the mask of hidden digits

Default: `blue.mix(gray)`

solution-color `color`

Color of the hidden digits if solution = true

Default: red

type-mask `str`

How the digits are hidden : "rect" for a round rectangle or "." for dots

Default: "rect"

list `array`

list of the digits to hide

Default: ()

solution `boolean`

show the carries and the hidden digits

Default: false

sign `boolean` or `content`

Sign of equality, false or a symbol \$ = \$

Default: false

space `length` or `relative length` or `fraction`

Space if more than one calculation on a line, default : 1fr

Default: 1fr

text-size `length` or `relative length`

size of the text : 1em

Default: 1em

..args `array` or `arguments`

Numbers to add

base-to-base

conversion from a base (default : 2) to another (default : 10).

```
// Conversion between bases (16 to 2):
#base-to-base("1A", base-from: 16, base-
to: 2, equal: true, fr: false)
```

$$(1A)_{16} = (11010)_2$$

```
#base-to-base("77", base-from: 8, base-
to: 2, equal: true, mode:"line", fr:
false)
```

$$\overline{77}^8 = \overline{111111}^2$$

```
// Full explanation (Base 16 to Base 8):
#base-to-base("1A.F", base-from: 16, base-to: 8, explain: true,mode: "line", fr: false)
```

1. Conversion from base 16 to base 10:

$$\overline{1A.F}^{16} = 1 \times 16^1 + 10 \times 16^0 + 15 \times 16^{-1} = \overline{26.9375}^{10}$$

2. Conversion from base 10 to base 8:

Integer part (successive divisions by 8):

- $26 = 8 \times 3 + 2$
- $3 = 8 \times 0 + 3$

Fractional part (successive multiplications by 8):

- $0.9375 \times 8 = 7.5$ (integer part: 7)
- $0.5 \times 8 = 4$ (integer part: 4)

Result: $\overline{1A.F}^{16} = \overline{32.74}^8$

Parameters

```
base-to-base(
  val: int float str,
  base-from: int,
  base-to: int,
  explain: boolean,
  equal: boolean,
  precision: int,
  fr: boolean,
  mode: str
) -> str int float
```

val int or float or str
number to convert

base-from `int`

original base

Default: `2`

base-to `int`

end base

Default: `10`

explain `boolean`

to get explanation of the conversion

Default: `false`

equal `boolean`

to get the conversion equality

Default: `false`

precision `int`

to limit the number of digits if the fractional part is infinite

Default: `6`

fr `boolean`

mode fr to get “,” instead of “.”

Default: `false`

mode `str`

a few modes to show the bases “paren”, “line”, “simple”

Default: `"paren"`

division

Divide decimal number a by b.

With extradigits, computes additional digits after the decimal point beyond those already present in a (when `cycle:false`). These extra digits are displayed in gray (`mode:"g"` by default), white (`mode:"0"`), as dots (`mode:"p"` or `"gp"`), or omitted (`mode:""`). Use `s:true/false` to show or hide the subtraction steps. You can highlight the repeating cycle in the quotient using `cycle:true`

(underlined with cycle-color), display it in parentheses with cycle:"", or display it in color with cycle:red. Fill-in-the-blank divisions are also supported with hide-result, list, and solution (using the type-mask, border-color, and solution-color parameters).

Examples:

```
#division(12.6, 3.75)
```

$$\begin{array}{r} 1260 \overline{) 375} \\ -1125 \\ \hline 135 \end{array}$$

```
// you can get the cycle of repeating
digits automatically and hide the
subtractions:
```

```
#division(121.5, 7, cycle:true, s:
false, fr: false)
```

$$\begin{array}{r} 17.3571428 \\ 7 \overline{) 121.5000000} \\ \underline{51} \\ 25 \\ \underline{40} \\ 50 \\ \underline{10} \\ 30 \\ \underline{20} \\ 60 \\ \underline{4} \end{array}$$

```
#division(121.5, 7,
cycle:true, fr:false, dx:.5em, dy: -.9em)
```

$$\begin{array}{r} 17.3571428 \\ 7 \overline{) 121.5000000} \\ \underline{-7} \\ 51 \\ \underline{-49} \\ 25 \\ \underline{-21} \\ 40 \\ \underline{-35} \\ 50 \\ \underline{-49} \\ 10 \\ \underline{-7} \\ 30 \\ \underline{-28} \\ 20 \\ \underline{-14} \\ 60 \\ \underline{-56} \\ 4 \end{array}$$

```
// in base 16 with a floating result
and arrows to show when digits are
pulled down:
```

```
#division(base: 16, extradigits: 2,
"1A8", "B", arrows: true, fr: false)
```

$$\begin{array}{r} 26.8B \\ B \overline{) 1A8.00} \\ \underline{-16} \downarrow \\ 48 \downarrow \\ \underline{-42} \downarrow \\ 60 \downarrow \\ \underline{-58} \downarrow \\ 80 \downarrow \\ \underline{-79} \\ 7 \end{array}$$

Parameters

```
division(  
  a: int float ,  
  b: int float ,  
  base: int ,  
  show-base: boolean ,  
  mode: str ,  
  extradigits: int ,  
  s: boolean ,  
  cycle: boolean str color ,  
  cycle-color: color ,  
  fr: boolean ,  
  size: length relative length ,  
  dx: length relative length ,  
  dy: length relative length ,  
  hide-result: boolean ,  
  list: array ,  
  solution: boolean ,  
  type-mask: str ,  
  border-color: color ,  
  solution-color: color ,  
  space: length relative length fraction  
) -> content
```

a int or float

dividend

b int or float

divisor

base int

Base (2 to 36)

Default: 10

show-base boolean

to show in which base the numbers and operation are

Default: false

mode str

Display mode for zeros added to the dividend: “g” (gray), “0” (black), “p” (small dot), “gp” (large dot), “” (empty)

Default: "g"

extradigits `int`

Number of additional quotient digits to compute

Default: `0`

s `boolean`

Show or hide the subtraction steps

Default: `true`

cycle `boolean` or `str` or `color`

Detect and display the repeating cycle automatically

Default: `false`

cycle-color `color`

Color of the cycle or underline

Default: `luma(50%)`

fr `boolean`

If `fr:true`, the decimal separator is “,” otherwise “.”

Default: `true`

size `length` or `relative length`

Column width

Default: `.7em`

dx `length` or `relative length`

`dx` for the `)` in mode `fr:false`

Default: `.6em`

dy `length` or `relative length`

`dy` for the `)` in mode `fr:false`

Default: `-.82em`

hide-result `boolean`

Hide the working steps and quotient (exercise mode)

Default: `false`

list `array`

List of cell indices to hide (dividend, divisor, steps, quotient)

Default: `()`

solution `boolean`

Show or hide the solution inside the masks

Default: `false`

type-mask `str`

Mask type ("rect" or "line")

Default: `"rect"`

border-color `color`

Color of the mask border

Default: `blue.mix(gray)`

solution-color `color`

Color of the solution text

Default: `red`

space `length` or `relative length` or `fraction`

Spacing when several calculations are on the same line

Default: `1fr`

egalite-decimale

Write the equality corresponding to a decimal division.

`#egalite-decimale(126, 25)`

$126 \div 25 = 5.04$

Parameters

```
egalite-decimale(  
  a: int float ,  
  b: int float ,  
  s: str  
) -> content
```

a int or float

dividend

b int or float

divisor

s str

sign: "e" for =, else approx

Default: "e"

egalite-euclidienne

Write the equality corresponding to a Euclidean division.

```
#egalite-euclidienne(126, 27)
```

```
126 = 4 × 27 + 18
```

Parameters

```
egalite-euclidienne(  
  a: int float ,  
  b: int float  
) -> content
```

a int or float

dividend

b int or float

divisor

from-base

conversion to decimal from another base (default : 2).

```
#from-base("1AF",base:16, fr: false)
```

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Parameters

```
from-base(  
  val: int float str,  
  base: int,  
  fr: boolean,  
  show-base: boolean  
) -> str int float
```

val int or float or str

number to convert

base int

origin base

Default: 2

fr boolean

mode fr to get “,” instead of “.” as a decimal separator

Default: false

show-base boolean

show the base of the result

Default: false

multiplication

Lay out the multiplication of 2 integer or decimal numbers with the possibility of hiding intermediate rows using hide-result, hiding a list of selected digits, then displaying the solution, with various configurable colors.

Examples:

```
#multiplication(12.5, 3.75, fr: false)
```

```
      12.5  
×     3.75  
-----  
      625  
+   8750  
+ 37500  
-----  
= 46.875
```

```
#multiplication(1215, 142, list:
(3,12,19,36), type-mask: "line", fr:
false)
```

$$\begin{array}{r}
 \dots 215 \\
 \times \quad 1\dots 2 \\
 \hline
 24\dots 0 \\
 + 48600 \\
 + 121500 \\
 \hline
 172530
 \end{array}$$

```
// In base 8 with all the carries and
floating numbers:
#multiplication(base: 8, "7.4",
"3.2", carries: true, carries-color:
(red,blue), fr: false)
```

$$\begin{array}{r}
 \overset{\text{red } 1}{7}.4 \\
 \times \quad 3.2 \\
 \hline
 170 \\
 + 2640 \\
 \hline
 \text{blue } +1 \text{ blue } +1 \\
 = 30.30
 \end{array}$$

```
// with only the multiplication
carries:
#multiplication(carries: "mult", 784,
53, fr: false, carries-color: teal)
```

$$\begin{array}{r}
 \overset{\text{teal } 2}{7} \overset{\text{teal } 1}{8} 4 \\
 \times \quad 53 \\
 \hline
 2352 \\
 + 39200 \\
 \hline
 = 41552
 \end{array}$$

```
// in binary with only the addition
carries:
#multiplication(base: 2, carries:
"add", "1011", "1101", fr: false)
```

$$\begin{array}{r}
 \times \quad 1011 \\
 \hline
 1101 \\
 + \quad 00 \\
 + \quad 101100 \\
 + \quad 1011000 \\
 \hline
 \text{red } +1 \text{ red } +1 \text{ red } +1 \text{ red } +1 \\
 = 10001111
 \end{array}$$

Parameters

```
multiplication(  
  a: int float str,  
  b: int float str,  
  base: int,  
  show-base: boolean,  
  carries: boolean str,  
  carries-color: color array,  
  mode: str,  
  sym: boolean str,  
  fr: boolean str,  
  size: length relative length,  
  hide-result: boolean,  
  list: array,  
  solution: boolean,  
  type-mask: str,  
  border-color: color,  
  solution-color: color,  
  space: length relative length fraction  
) -> content
```

a int or float or str

first factor

b int or float or str

second factor

base int

Base (2 to 36)

Default: 10

show-base boolean

to show in which base the numbers and operation are

Default: false

carries boolean or str

to show the carries : false, true (“all”), “mult”, “add”

Default: false

carries-color `color` or array

Color of the carries

Default: red

mode `str`

Display mode for shifted rows: “g” (default) for gray zeros, “0” for normal zeros, “p” for small dots, “gp” for larger dots, “” or anything else for nothing

Default: “g”

sym `boolean` or `str`

Show the addition “+” signs and the final “=” if true; only “+” if “+”, only “=” if “=”, otherwise nothing

Default: true

fr `boolean` or `str`

If fr:true, the decimal separator is “,” otherwise “.”

Default: true

size `length` or `relative length`

Column width

Default: 1em

hide-result `boolean`

Hide intermediate rows and the final result for exercises

Default: false

list array

List of cell indices to hide (fill-in-the-blank multiplication)

Default: ()

solution `boolean`

Show or hide the solution inside the masks

Default: false

type-mask str

Mask type ("rect" or "line")

Default: "rect"

border-color color

Color of the mask border

Default: blue.mix(gray)

solution-color color

Color of the solution text

Default: red

space length or relative length or fraction

Spacing when several calculations are on the same line

Default: 1fr

racine

Compute the square root of a, with extradigits additional decimal places, optionally showing digit pairs using groups in groups-color, displaying the construction step by step with step, hiding subtraction steps with s, and hiding the result or a list of digits.

Examples:

```
#racine(24368,size: 1em,list: (2,27))
```

$$\begin{array}{r|l} 24\text{ } \textcolor{blue}{6}8 & 1\text{ } \textcolor{blue}{6} \\ \hline -1 & 1^2 = 1 < 2 \\ \textcolor{red}{1}43 & 25 \times \textcolor{blue}{5} = 125 \\ -125 & 306 \times \textcolor{blue}{6} = 1836 \\ \hline \textcolor{red}{1}868 & \\ -1836 & \\ \hline 32 & \end{array}$$

```
#grid(columns: 3*(1fr,), row-gutter: 1em,
  ..for i in range(1,11) {[
    ==== Étape #i
    #racine(24368, extradigits: 1,groups: true, step: i, fr: false)
  ],)+if i == 1 {[[],[],)}]}
)
```

Étape 1

$$\begin{array}{r|l} \overline{24368} & 1 \\ -\frac{1}{1} & 1^2 = 1 < 2 \end{array}$$

Étape 2

$$\begin{array}{r|l} \overline{24368} & 1 \\ -\frac{1}{143} & 1^2 = 1 < 2 \\ & 2c \times c \leq 143 \end{array}$$

Étape 3

$$\begin{array}{r|l} \overline{24368} & 1 \\ -\frac{1}{143} & 1^2 = 1 < 2 \\ & 25 \times 5 = 125 \end{array}$$

Étape 4

$$\begin{array}{r|l} \overline{24368} & 15 \\ -\frac{1}{143} & 1^2 = 1 < 2 \\ -\frac{125}{18} & 25 \times 5 = 125 \end{array}$$

Étape 5

$$\begin{array}{r|l} \overline{24368} & 15 \\ -\frac{1}{143} & 1^2 = 1 < 2 \\ -\frac{125}{1868} & 25 \times 5 = 125 \\ & 30c \times c \leq 1868 \end{array}$$

Étape 6

$$\begin{array}{r|l} \overline{24368} & 15 \\ -\frac{1}{143} & 1^2 = 1 < 2 \\ -\frac{125}{1868} & 25 \times 5 = 125 \\ & 306 \times 6 = 1836 \end{array}$$

Étape 7

$$\begin{array}{r|l} \overline{24368} & 156 \\ -\frac{1}{143} & 1^2 = 1 < 2 \\ -\frac{125}{1868} & 25 \times 5 = 125 \\ -\frac{1836}{32} & 306 \times 6 = 1836 \end{array}$$

Étape 8

$$\begin{array}{r|l} \overline{24368.00} & 156 \\ -\frac{1}{143} & 1^2 = 1 < 2 \\ -\frac{125}{1868} & 25 \times 5 = 125 \\ -\frac{1836}{3200} & 306 \times 6 = 1836 \\ & 312c \times c \leq 3200 \end{array}$$

Étape 9

$$\begin{array}{r|l} \overline{24368.00} & 156 \\ -\frac{1}{143} & 1^2 = 1 < 2 \\ -\frac{125}{1868} & 25 \times 5 = 125 \\ -\frac{1836}{3200} & 306 \times 6 = 1836 \\ & 3121 \times 1 = 3121 \end{array}$$

Étape 10

$$\begin{array}{r|l} \overline{24368.00} & 156.1 \\ -\frac{1}{143} & 1^2 = 1 < 2 \\ -\frac{125}{1868} & 25 \times 5 = 125 \\ -\frac{1836}{3200} & 306 \times 6 = 1836 \\ -\frac{3121}{79} & 3121 \times 1 = 3121 \end{array}$$

Parameters

```
racine(  
  a: int float ,  
  extradigits: int ,  
  groups: boolean ,  
  groups-color: color ,  
  step: none int ,  
  step-color: color ,  
  s: boolean ,  
  fr: boolean ,  
  size: length relative length ,  
  hide-result: boolean ,  
  list: array ,  
  solution: boolean ,  
  type: str ,  
  border-color: color ,  
  solution-color: color ,  
  space: length relative length fraction ,  
  ..args: arguments  
) -> content
```

a int or float

number

extradigits int

Number of additional “00” digit pairs to compute after the decimal point

Default: 0

groups boolean

Show brackets above each pair of digits

Default: false

groups-color color

Color of the brackets

Default: blue.darken(20%)

step none or int

Step to display in the step-by-step construction

Default: none

step-color `color`

Color of the step indicator

Default: `blue.darken(20%)`

s `boolean`

Show or hide the subtraction steps below the dividend

Default: `true`

fr `boolean`

If `fr:true`, the decimal separator is “,” otherwise “.”

Default: `true`

size `length` or `relative length`

Column width

Default: `0.65em`

hide-result `boolean`

Hide the construction steps, operations, and result (exercise mode)

Default: `false`

list `array`

List of cell indices to hide (fill-in-the-blank exercise)

Default: `()`

solution `boolean`

Show or hide the solution inside the masks

Default: `false`

type `str`

Mask type (“rect” or “line”)

Default: `"rect"`

border-color `color`

Color of the mask border

Default: `blue.mix(gray)`

solution-color `color`

Color of the solution text

Default: `red`

space `length` or `relative length` or `fraction`

Right spacing

Default: `1fr`

..args `arguments`

Other arguments to the square root, ex:[...]

sous-en-ligne

Compute the difference of all the supplied numbers inline.

```
$#sous-en-ligne(198.7,107.3,-35)$
```

```
198.7 - 107.3 - (-35) = 126.4
```

Parameters

```
sous-en-ligne(  
  ..ops: arguments ,  
  digits: auto int  
) -> content
```

..ops `arguments`

list of numbers

soustraction

Parameters

```
soustraction(  
  base: int,  
  show-base: boolean,  
  show-borrow: boolean,  
  canceled: boolean,  
  fr: boolean,  
  carries-color: color,  
  size: length relative length,  
  zeros: boolean,  
  hide-result: boolean,  
  border-color: color,  
  solution-color: color,  
  type-mask: str,  
  list: array,  
  solution: boolean,  
  sign: boolean content,  
  text-size: length relative length,  
  space: length relative length fraction,  
  ..args: array arguments  
) -> content
```

base int

Base (from 2 to 36)

Default: 10

show-base boolean

to show the base of the numbers and operation

Default: false

show-borrow boolean

Show or hide the borrows

Default: true

canceled boolean

If fr:false, show or hide the crossed-out original digit

Default: false

fr boolean

French notation or English borrowing method

Default: true

carries-color `color`

Color of the borrows

Default: red

size `length` or `relative length`

Column width

Default: `1em`

zeros `boolean`

Add trailing zeros when needed

Default: `true`

hide-result `boolean`

Hide the borrows and the result

Default: `false`

border-color `color`

Color of the border around hidden digits

Default: `blue.mix(gray)`

solution-color `color`

Color of hidden digits when solution = true

Default: red

type-mask `str`

How hidden digits are displayed: "rect" for a rounded rectangle or ...

Default: `"rect"`

list `array`

List of digits to hide in a fill-in-the-blank subtraction

Default: `()`

solution `boolean`

Show the borrows and the result in hide-result mode (for answer keys)

Default: `false`

sign `boolean` or `content`

Equality sign, false or a symbol \$ = \$

Default: `false`

text-size `length` or `relative length`

Text size: 1em

Default: `1em`

space `length` or `relative length` or `fraction`

Spacing when several calculations are on the same line

Default: `1fr`

..args `array` or `arguments`

Numbers

to-base

conversion from decimal to another base (default : 2).

```
#to-base(base:16,"7.8",fr: false)
```

7.CCCCCC

```
#to-base(100, base: 8, fr: false)
```

144

Parameters

```
to-base(  
  val: int float str,  
  base: int,  
  fr: boolean,  
  show-base: boolean,  
  precision: int  
) -> str int float
```

val `int` or `float` or `str`

number to convert

base `int`

base to convert to

Default: `2`

fr `boolean`

mode fr to get “,” instead of “.” as a decimal separator

Default: `false`

show-base `boolean`

show the base of the result

Default: `false`

precision `int`

to limit the number of digits if the fractional part is infinite

Default: `6`

Additions and subtractions of durations

- `addition-durees()`
- `prepa-duree()`
- `soustraction-durees()`

addition-durees

Sets up the addition of two or more durations.

You need to change units: ([j], [h], [min], [s]) with units: ([d], [h], [min], [s]).

Each duration is an array in one of the following forms:

- (days, hours, minutes, seconds).
- (hours, minutes, seconds).
- (minutes, seconds).

Examples:

```
#addition-durees(  
  (1, 2, 45, 50),  
  (0, 1, 20, 30),  
)
```

	j	h	min	s
		1	1	
	1	02	45	50
+	0	01	20	30
	1	04	06	20

```
#addition-durees(  
  (  
    (1, 2, 45, 50),  
    (0, 1, 20, 30),  
    (2, 23, 55, 45),  
  ),  
  hide-result: true,  
)
```

	j	h	min	s
	1	2	2	
	1	02	45	50
+	0	01	20	30
+	2	23	55	45

Parameters

```
addition-durees(  
  show-carry: boolean ,  
  carries-color: color ,  
  carry-size: length relative length ,  
  size: length relative length ,  
  conversion: boolean ,  
  hide-empty: boolean ,  
  hide-result: boolean ,  
  border-color: color ,  
  solution-color: color ,  
  type-mask: str ,  
  list: array ,  
  solution: boolean ,  
  sign: boolean content ,  
  text-size: length relative length ,  
  show-units: boolean ,  
  units: array ,  
  units-size: length relative length ,  
  units-color: color ,  
  zero-pad: boolean ,  
  align: boolean ,  
  verticals: boolean ,  
  ..args: array arguments  
) -> content
```

show-carry boolean

Show carries.

Default: **true**

carries-color color

Color of the carries.

Default: red

carry-size length or relative length

Size of the carries and the + and = symbols, max. 1em.

Default: **.8em**

size length or relative length

Width of the columns corresponding to the units.

Default: **2.5em**

conversion `boolean`

Convert the result instead of displaying carries.

Default: `false`

hide-empty `boolean`

Hide columns containing only zeros.

Default: `true`

hide-result `boolean`

Hide the result.

Default: `false`

border-color `color`

Border color of hidden values.

Default: `blue.mix(gray)`

solution-color `color`

Color of values in the solution.

Default: `red`

type-mask `str`

Mask type: “rect” or “line”.

Default: `"rect"`

list `array`

Indices of the cells to hide.

Default: `()`

solution `boolean`

Show the solution for hidden cells.

Default: `false`

sign `boolean` or `content`

Sign placed before the result.

Default: `false`

text-size `length` or `relative length`

Text size.

Default: `1em`

show-units `boolean`

Show unit names.

Default: `true`

units `array`

Names of the four units.

Default: `([d], [h], [min], [s])`

units-size `length` or `relative length`

Unit size, max. 1em.

Default: `.75em`

units-color `color`

Color of the units.

Default: `gray`

zero-pad `boolean`

Display hours, minutes, and seconds using two digits.

Default: `true`

align `boolean`

Align the digits of the units.

Default: `true`

verticals `boolean`

Vertical dotted lines to separate the columns.

Default: `false`

..args `array` or `arguments`

List of durations

prepa-duree

Converts decimal or fractional durations into integer durations

Parameters

`prepa-duree(..args: array arguments) -> array`

..args `array` or `arguments`

List of 2 to 4 numbers (d, h, min, s) to (min, s)

soustraction-durees

Sets up the subtraction of two durations.

You need to change units: ([j], [h], [min], [s]) with units: ([d], [h], [min], [s]).

Each duration is an array in one of the following forms:

- (days, hours, minutes, seconds);
- (hours, minutes, seconds);
- (minutes, seconds).

The first duration must be greater than or equal to the second.

Examples:

```
#soustraction-durees(  
  (2, 4, 15, 10),  
  (1, 6, 45, 30),  
)
```

	j	h	min	s
		⁺²⁴	⁺⁶⁰	⁺⁶⁰
	2	04	15	10
-	1	06	45	30
	⁺¹	⁺¹	⁺¹	
	0	21	29	40

```
#soustraction-durees(
  (2, 15, 10),
  (1, 45, 30),
  hide-result: true,
)
```

	h	min	s
	2	15	10
–	1	45	30

Parameters

```
soustraction-durees(
  show-borrow: boolean,
  carries-color: color,
  carry-size: length relative length,
  size: length relative length,
  conversion: boolean,
  hide-empty: boolean,
  hide-result: boolean,
  border-color: color,
  solution-color: color,
  type-mask: str,
  list: array,
  solution: boolean,
  sign: boolean content,
  text-size: length relative length,
  show-units: boolean,
  units: array,
  units-size: length relative length,
  units-color: color,
  zero-pad: boolean,
  align: boolean,
  verticals: boolean,
  ..args: array arguments
) -> content
```

show-borrow boolean

Show borrows and carries.

Default: true

carries-color color

Color of the borrows and carries.

Default: red

carry-size length or relative length

Size of the carries and the + and = symbols, max. 1em.

Default: .7em

size `length` or `relative length`

Width of the columns.

Default: `2.5em`

conversion `boolean`

Convert the minuend instead of displaying carries.

Default: `false`

hide-empty `boolean`

Hide columns containing only zeros.

Default: `true`

hide-result `boolean`

Hide the result.

Default: `false`

border-color `color`

Border color of hidden values.

Default: `blue.mix(gray)`

solution-color `color`

Color of values in the solution.

Default: `red`

type-mask `str`

Mask type: “rect” or “line”.

Default: `"rect"`

list `array`

Indices of the cells to hide.

Default: `()`

solution `boolean`

Show the solution.

Default: `false`

sign `boolean` or `content`

Sign placed before the result.

Default: `false`

text-size `length` or `relative length`

Text size.

Default: `1em`

show-units `boolean`

Show unit names.

Default: `true`

units `array`

Names of the four units.

Default: (`[d]`, `[h]`, `[min]`, `[s]`)

units-size `length` or `relative length`

Unit size, max. 1em.

Default: `.75em`

units-color `color`

Color of the units.

Default: `gray`

zero-pad `boolean`

Display hours, minutes, and seconds using two digits.

Default: `true`

align `boolean`

Align the digits of the units.

Default: `true`

verticals `boolean`

Vertical dotted lines to separate the columns.

Default: `false`

..args `array` or `arguments`

List of durations

Step-by-step calculations and highlighting

- `detail()`
- `etapes-calcul()`

detail

Function for writing calculation steps manually.

```
#detail("A=15/21=(5*r3)/  
(7*r3)=5/7",add-  
rounded:true,c:"circle")
```

$$A = \frac{15}{21} = \frac{5 \times \textcircled{3}}{7 \times \textcircled{3}} = \frac{5}{7} \approx 0,71$$

```
#detail("B=(15s2)/(21s18)=(5*r3 s2)/  
(7*r3 s(9*2))=(5 ps2)/(21 bs2)  
=5/21",c:"color",vertical:true, add-  
rounded: true)
```

$$\begin{aligned} B &= \frac{15\sqrt{2}}{21\sqrt{18}} \\ &= \frac{5 \times \textcolor{red}{3}\sqrt{2}}{7 \times \textcolor{red}{3}\sqrt{9 \times 2}} \\ &= \frac{5\sqrt{2}}{21\sqrt{2}} \\ &= \frac{5}{21} \\ &\approx 0,24 \end{aligned}$$

```
#detail("cal(A)=pi * r^2 = pi*5^2=25  
pi",add-rounded: true,digits: 4)
```

$$\mathcal{A} = \pi \times r^2 = \pi \times 5^2 = 25\pi \approx 25 \times 3,1416 \approx 78,54 \approx 78,54$$

Parameters

```
detail(  
  expr-str: str,  
  vertical: boolean,  
  add-rounded: boolean,  
  digits: int,  
  appro: int,  
  todo: boolean,  
  space: length relative length fraction,  
  a: color,  
  b: color,  
  f: color,  
  g: color,  
  r: color,  
  l: color,  
  m: color,  
  n: color,  
  o: color,  
  t: color,  
  p: color,  
  y: color,  
  c: str  
) -> content
```

expr-str `str`

The detailed calculation expression

vertical `boolean`

Vertical layout

Default: `false`

add-rounded `boolean`

Append a rounded value at the end

Default: `false`

digits `int`

Number of decimal places for the rounded value

Default: `2`

appro `int`

Step from which “=” is replaced by “≈” when appropriate

Default: `0`

todo `boolean`

If `todo: true`, only the initial expression is displayed

Default: `false`

space `length` or `relative length` or `fraction`

Spacing before the next calculation

Default: `1fr`

a `color`

a: aqua; can be changed from Typst’s default color if needed

Default: aqua

b `color`

b: blue; can be changed from Typst’s default color if needed

Default: blue

f 

f: fuchsia; can be changed from Typst's default color if needed

Default: fuchsia

g 

g: green; can be changed from Typst's default color if needed

Default: green

r 

r: red; can be changed from Typst's default color if needed

Default: red

l 

l: gray; can be changed from Typst's default color if needed

Default: gray

m 

m: maroon; can be changed from Typst's default color if needed

Default: maroon

n 

n: rgb(8,111,189) (navy); can be changed if needed

Default: `rgb("#086fbd")`

o 

o: orange; can be changed from Typst's default color if needed

Default: orange

t 

t: rgb(0,128,128) (teal); can be changed if needed

Default: `rgb(0,128,128)`

p color

p: purple; can be changed from Typst's default color if needed

Default: purple

y color

y: yellow; can be changed from Typst's default color if needed

Default: yellow

c str

Display mode: "cancel", "circle", or "color"

Default: "color"

etapes-calcul

Show the step-by-step evaluation of `expr-str` (unless `todo: true`), with the option to perform independent operations of the same precedence simultaneously using `concomitant`, display the work vertically or horizontally, highlight the current step (`highlight`), and use fraction mode to keep computations exact by forcing calculations to continue with fractions.

```
#etapes-calcul("[ 6 + 5 * (23 + 8 /  
2) ] * 8", fraction: false, vertical:  
true, n: 0)
```

$$\begin{aligned} & \left[6 + 5 \times \left(23 + \frac{8}{2} \right) \right] \times 8 \\ &= [6 + 5 \times (23 + 4)] \times 8 \\ &= [6 + 5 \times 27] \times 8 \\ &= [6 + 135] \times 8 \\ &= 141 \times 8 \\ &= 1128 \end{aligned}$$

```
#etapes-calcul("(200 - 45 * 2) / ((4 *  
7 - 3 : 6) * 2)", concomitant: true)
```

$$\frac{200 - 45 \times 2}{(4 \times 7 - 3 \div 6) \times 2} = \frac{200 - 90}{(28 - 0,5) \times 2} = \frac{110}{27,5 \times 2} = \frac{110}{55} = 2$$

```
#etapes-calcul("2 * pi * 3", fraction:  
false, name: "B", n: 3)
```

$$B = 2 \times \pi \times 3 = 2\pi \times 3 = 6\pi \approx 18,85$$

```
#etapes-calcul("ln(exp(1)) + 3",  
fraction: true, name: "E")
```

$$E = \ln(\exp(1)) + 3 = 1 + 3 = 4$$

```
#etapes-calcul("sqrt(8)*sqrt(27)",  
fraction: true)
```

$$\sqrt{8} \times \sqrt{27} = 2\sqrt{2} \times \sqrt{27} = 2\sqrt{2} \times 3\sqrt{3} = 6\sqrt{6}$$

Parameters

```
etapes-calcul(  
  expr-str: str,  
  mode: str,  
  concomitant: boolean,  
  vertical: boolean,  
  digits: int,  
  n: int,  
  name: str content,  
  todo: boolean,  
  space: length relative length fraction,  
  fraction: boolean,  
  highlight: color boolean none,  
  skip: array  
) -> content
```

expr-str str

expression to calculate

mode str

“fr” or “en” mode for the decimal separator

Default: "fr"

concomitant boolean

Set to true to perform independent operations of the same precedence simultaneously

Default: false

vertical boolean

Display the calculation vertically

Default: false

digits int

Number of decimal places for calculations involving π , etc.

Default: 2

n `int`

Number of steps (counted from the end) at which “=” is replaced by “≈” when appropriate

Default: `0`

name `str` or `content`

Label to prepend to the expression

Default: `""`

todo `boolean`

Exercise mode: only the initial expression followed by “=” is displayed

Default: `false`

space `length` or `relative length` or `fraction`

Horizontal spacing between two etapes-calcul outputs

Default: `1fr`

fraction `boolean`

In fraction mode, computations remain exact, either as fractions or simplified forms such as 5π

Default: `false`

highlight `color` or `boolean` or `none`

Either a color used to highlight the current operation, `false` for bold, or `none` for no highlighting

Default: `rgb("#1e88e5")`

skip `array`

List of steps to skip if needed

Default: `()`